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STATISTICS
Session 8: Classification & Prediction

Explanation vs Prediction

Explain

- The goal is to understand relationships between variables and uncover why something happens.
  - **Example:** Using regression to see how difference in ratings, training hours, coaching experience, etc., influence score margin.
  - **Focus:** coefficients, significance tests, causal or correlational interpretations.

Predict

- The goal is to forecast outcomes as accurately as possible, often without needing to interpret the underlying relationships.
  - **Example:** Using a machine learning model to predict exam scores for future students.
  - **Focus:** accuracy metrics (e.g., RMSE, accuracy, precision/recall), out-of-sample performance.

Recap: Multiple Regression

- Last time we looked at building regression models when the outcome was numeric

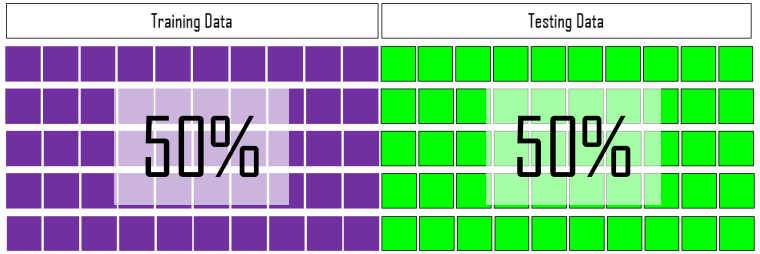
$$y = b_0 + b_1 x_1 + b_2 x_2 + \cdots b_k x_k + \epsilon$$

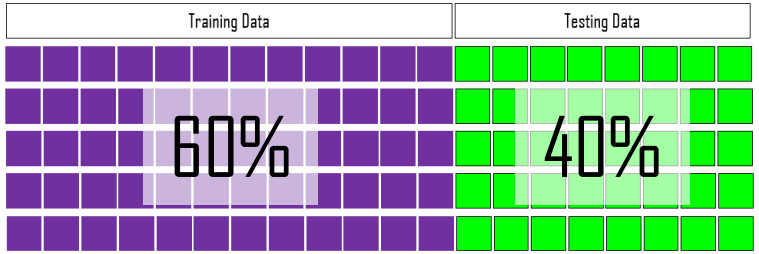
- For example, we could look at modelling:
  - Income ($)
  - Exam score (points)
  - Height (cm)
  - Return on investment (ROI in points)

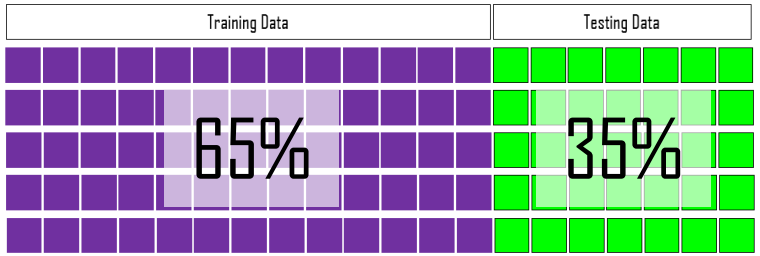
- The key is: the outcome (y) was numeric and measured on a continuous scale

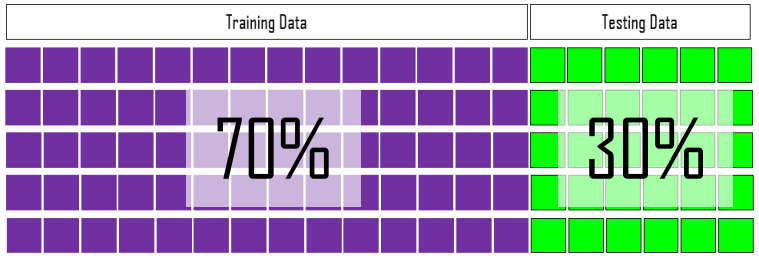
Training and Testing Data Sets

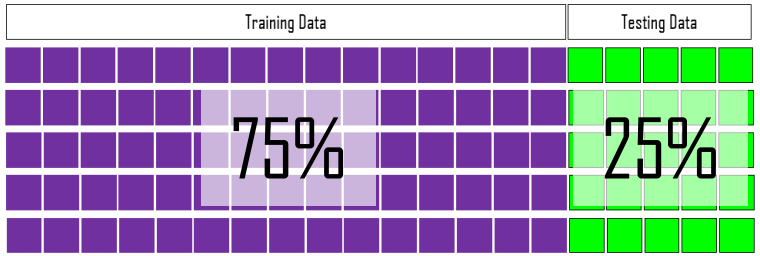
- **Training set:** used to build / fit the model. The model 'learns' the relationship between predictors (X) and the outcome (Y).
- **Testing set:** Used to evaluate the model's performance on unseen data, checking if it generalises beyond the training data.
- There is no universally correct training/test split.
  - A larger training set gives the model more observations to learn from.

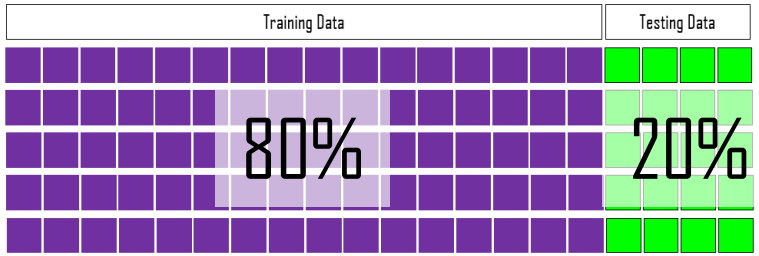








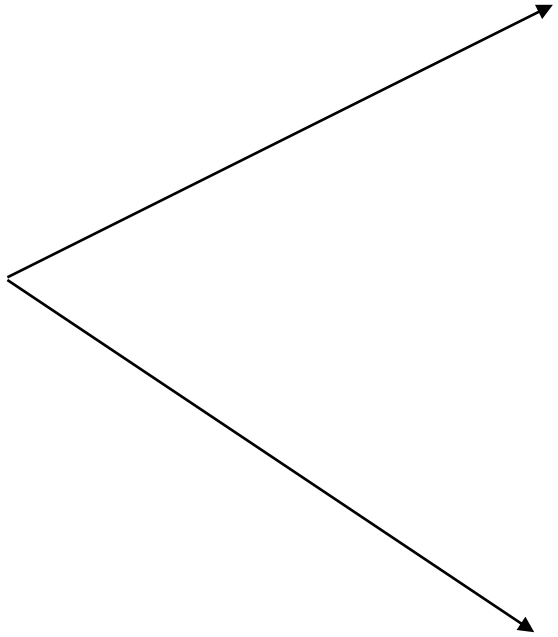


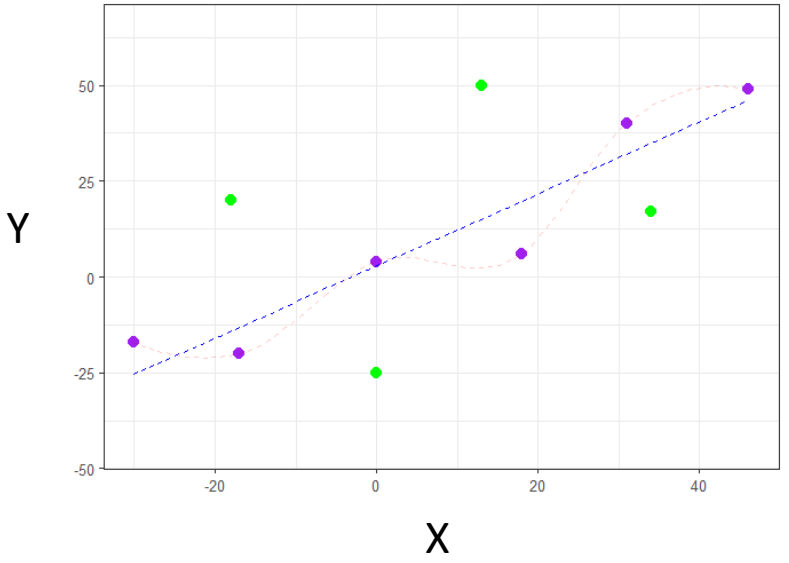


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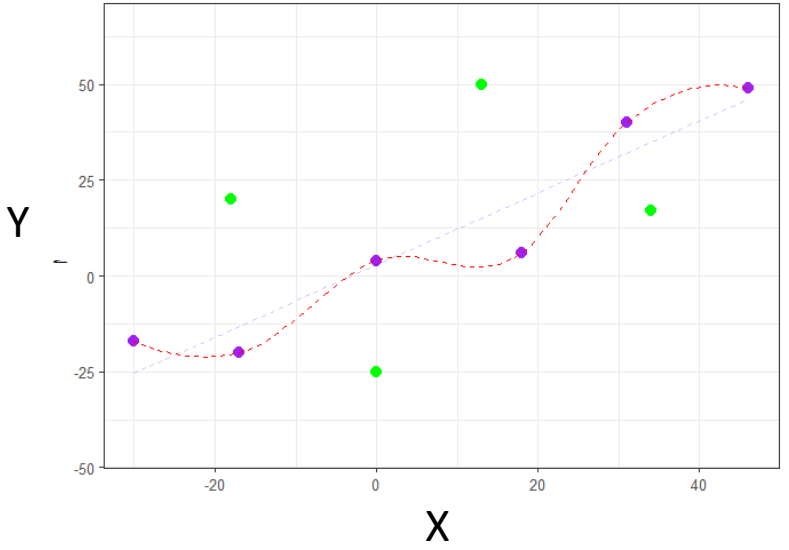
Comparing Models

X	Y
-17	-20
-30	-17
0	4
18	6
31	40
46	49





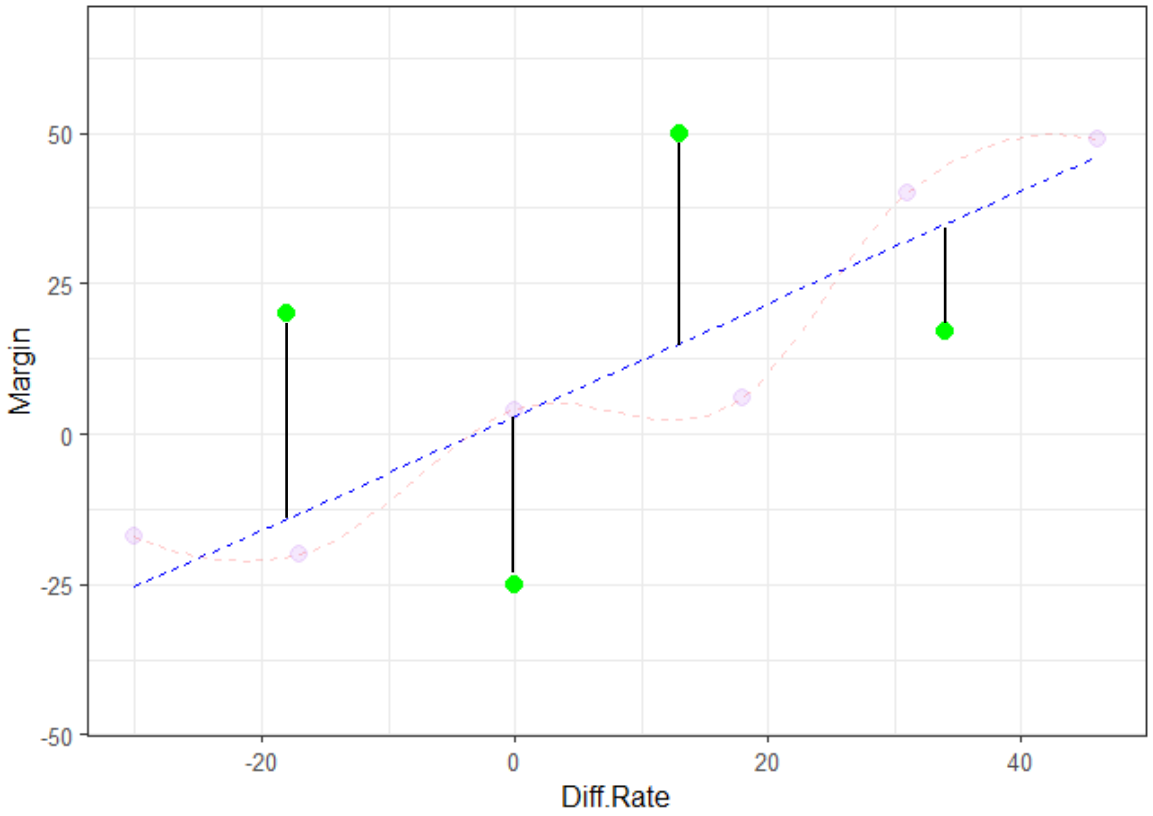
Model 1

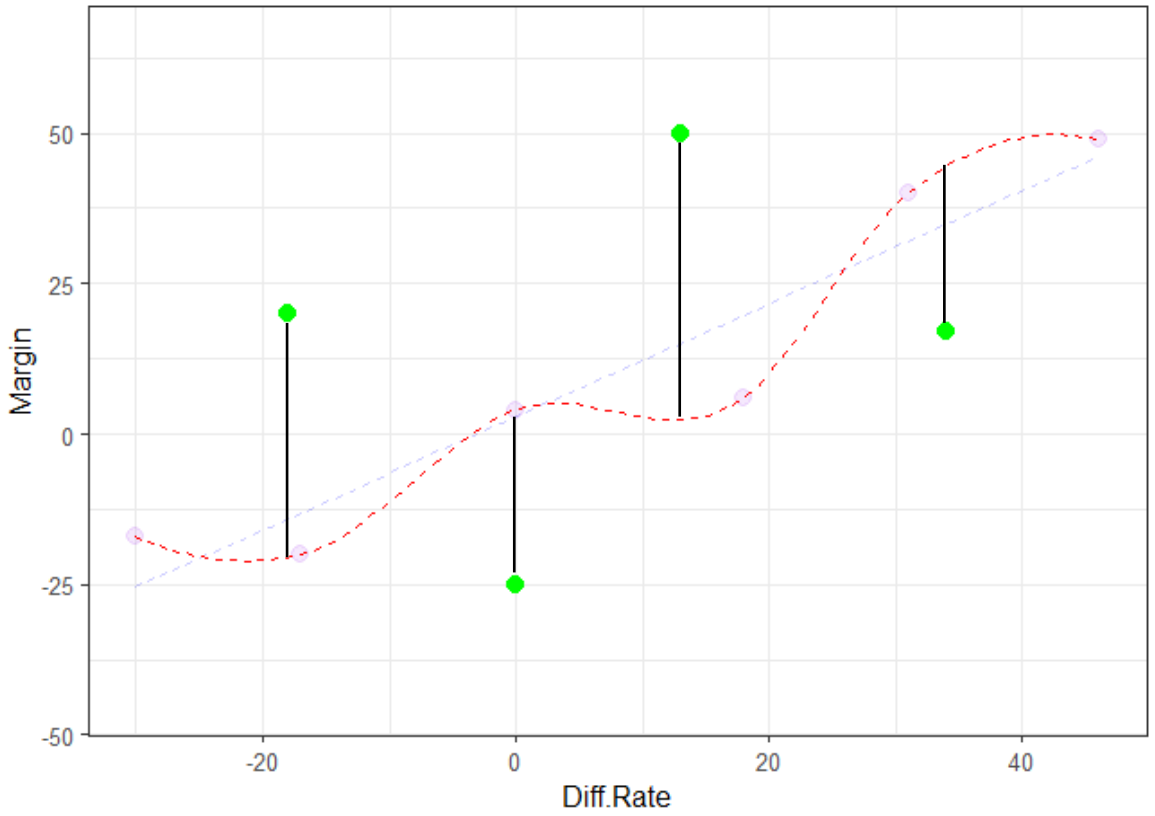


Model 2

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Comparing Models





New (Test) data is used to evaluate the models – we want the model that has less error

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Regression vs Classification

- Suppose that instead of modelling a continuous outcome (-∞, +∞), we wanted to examine a binary outcome (0, 1).
- We cannot use linear regression because the model will make predictions less than 0 and greater than 1.
- Therefore we will need a different type of model – one that fits a binary (0, 1) outcome.
- Sometimes, these types of models are called 'classification' models.

Continuous	Binary
-17	0
-30	0
0	1
18	1
31	0
46	1
10	0
5	1
20	1

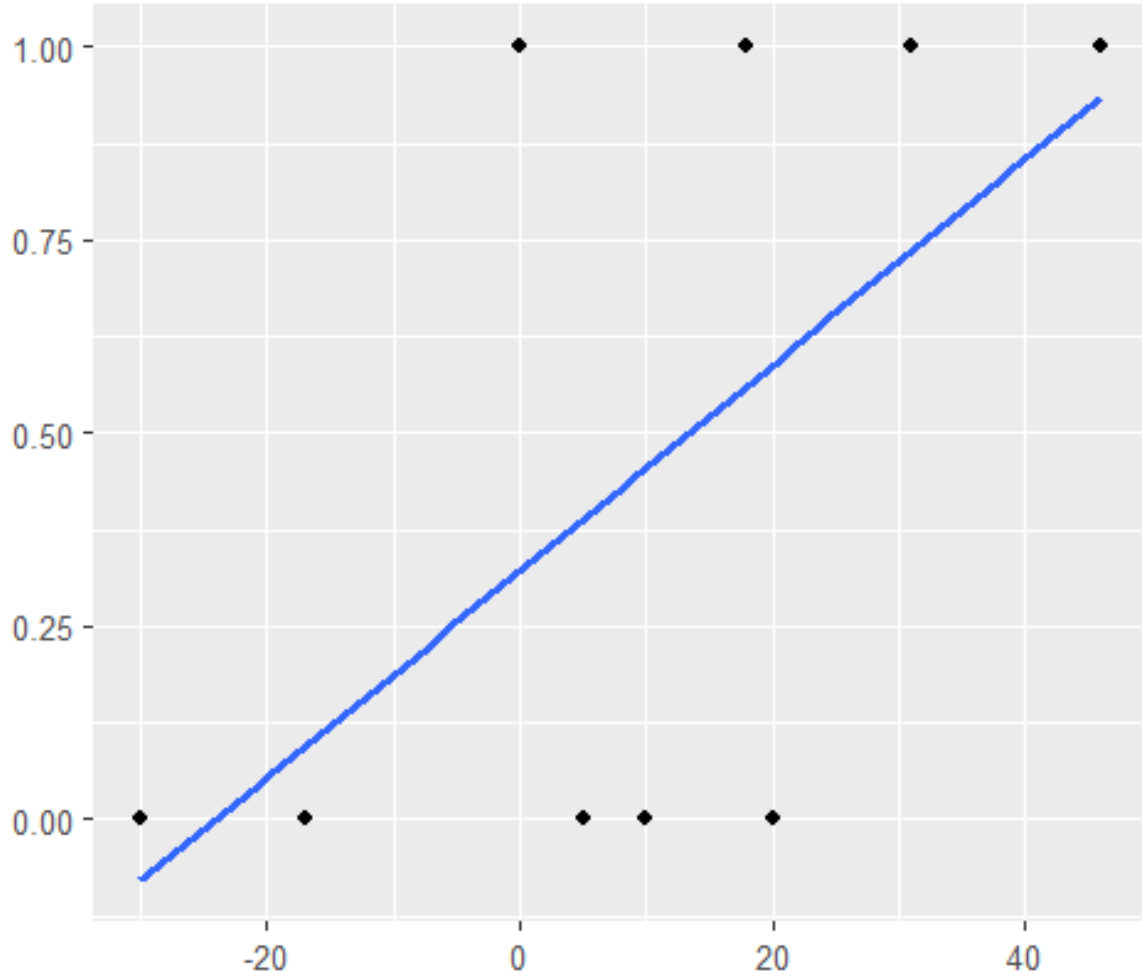
Regression vs Classification

- Suppose that instead of modelling a continuous outcome ($-\infty$, $+\infty$), we wanted to examine a binary outcome (0, 1).
- We cannot use linear regression because the model will make predictions less than 0 and greater than 1.
- Therefore we will need a different type of model – one that fits a binary (0, 1) outcome.
- Sometimes, these types of models are called 'classification' models.

Continuous	Binary
-17	0
-30	0
0	1
18	1
31	0
46	1
10	0
5	1
20	1

Linear Probability Model (LPM)

- The linear probability model (LPM) uses linear regression to model a binary outcome, coded as 0 or 1.
- Predicted values represent the probability that the outcome equals 1.
- Each coefficient describes the change in probability for a one-unit increase in a predictor, holding other predictors constant.
- A key limitation is that predicted probabilities can fall below 0 or above 1.



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Linear Probability Model (LPM)

$$P(Y = 1 \mid X) = \beta_0 + \beta_1 X$$

One-predictor form

1 Binary outcome

Use linear regression when Y is coded 0 or 1.

0

$Y = 0$
(e.g., no)

1

$Y = 1$
(e.g., yes)

2 Predict a probability

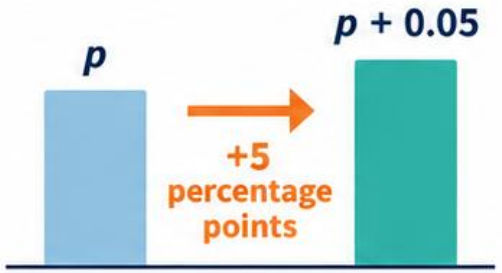
The fitted value estimates $P(Y = 1 \mid X)$.



Predicted probability
$P(Y = 1 \mid X)$

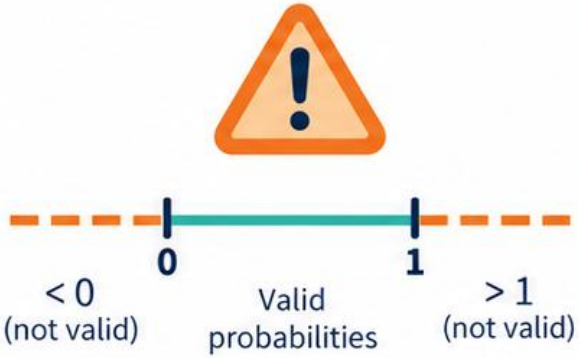
3 Interpret the slope

A coefficient of 0.05 means a 5 percentage point increase in probability for a one-unit increase in X, holding other predictors constant.



4 Main limitation

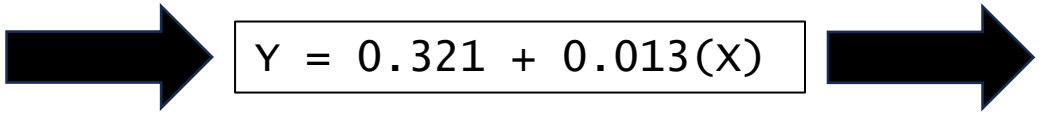
Predictions can fall below 0 or above 1.



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Linear Probability Model (LPM)

X	Y
-17	0
-30	0
0	1
18	1
31	1
46	1
10	0
5	0
20	0

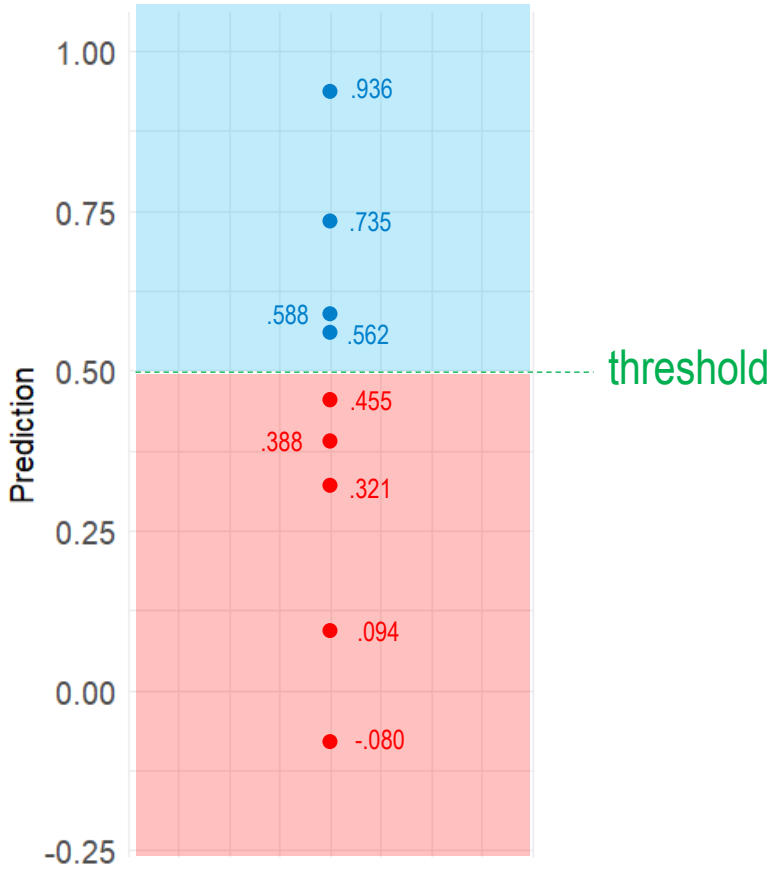


Prediction
0.321 + 0.013(-17) = 0.094
0.321 + 0.013(-30) = -0.080
0.321 + 0.013(0) = 0.321
0.321 + 0.013(18) = 0.562
0.321 + 0.013(31) = 0.735
0.321 + 0.013(46) = 0.936
0.321 + 0.013(10) = 0.455
0.321 + 0.013(5) = 0.388
0.321 + 0.013(20) = 0.588

Thresholds & Classification

$$Y = 0.321 + 0.013(X)$$

X	Y	Pred	Class
-17	0	.094	0
-30	0	-.080	0
0	1	.321	0
18	1	.562	1
31	1	.735	1
46	1	.936	1
10	0	.455	0
5	0	.388	0
20	0	.588	1



		ACTUAL		
		1	0	TOTAL
PREDICTED	1	3	1	4
	0	1	4	5
TOTAL		4	5	9

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Classification Errors

		ACTUAL	
		1	0
PREDICTED	1	3	1
	0	1	4

		ACTUAL	
		1	0
PREDICTED	1	True Positive	False Positive
	0	False Negative	True Negative

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Comparing Models - Classification Errors

Model 1

		ACTUAL	
		1	0
PREDICTED	1	242	47
	0	51	210

False Positive Rate (Type I Error)

$$= \frac{FP}{FP + TN}$$

$$= \frac{47}{47 + 210}$$

$$= 0.183$$

False Negative Rate (Type II Error)

$$= \frac{FN}{FN + TP}$$

$$= \frac{51}{51 + 242}$$

$$= 0.174$$

Model 2

		ACTUAL	
		1	0
PREDICTED	1	239	40
	0	32	141

False Positive Rate (Type I Error)

$$= \frac{FP}{FP + TN}$$

$$= \frac{40}{40 + 141}$$

$$= 0.221$$

False Negative Rate (Type II Error)

$$= \frac{FN}{FN + TP}$$

$$= \frac{32}{32 + 239}$$

$$= 0.118$$

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Comparing Models - Classification Errors

Model 1

		ACTUAL	
		1	0
PREDICTED	1	242	47
	0	51	210

$$\text{Overall Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{242 + 210}{242 + 210 + 51 + 47}$$

$$= 0.823$$

Model 2

		ACTUAL	
		1	0
PREDICTED	1	239	40
	0	32	141

$$\text{Overall Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

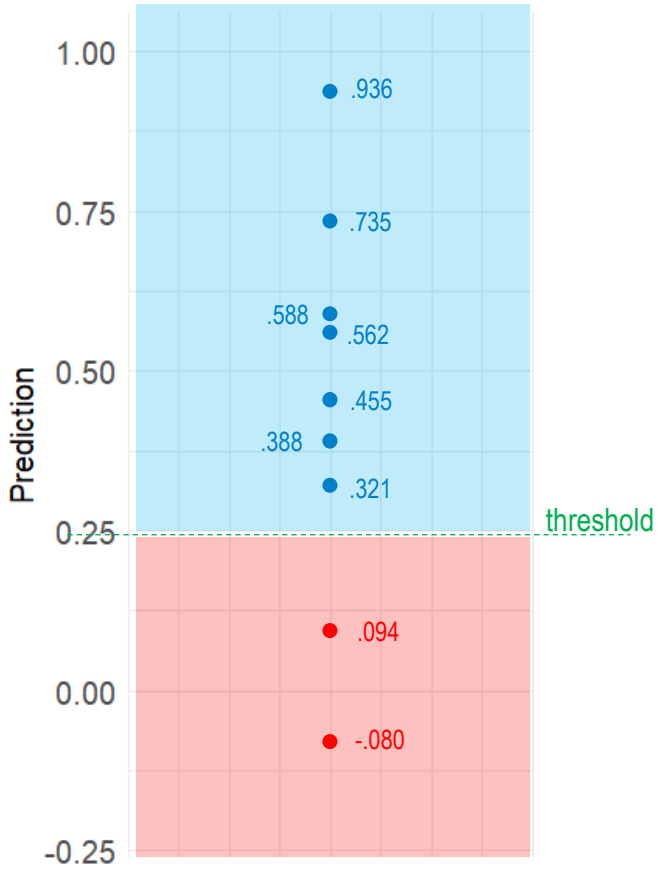
$$= \frac{239 + 141}{239 + 141 + 32 + 40}$$

$$= 0.841$$

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Using Different Thresholds

X	Y	Pred	Class
-17	0	.094	0
-30	0	-.080	0
0	1	.321	1
18	1	.562	1
31	1	.735	1
46	1	.936	1
10	0	.455	1
5	0	.388	1
20	0	.588	1



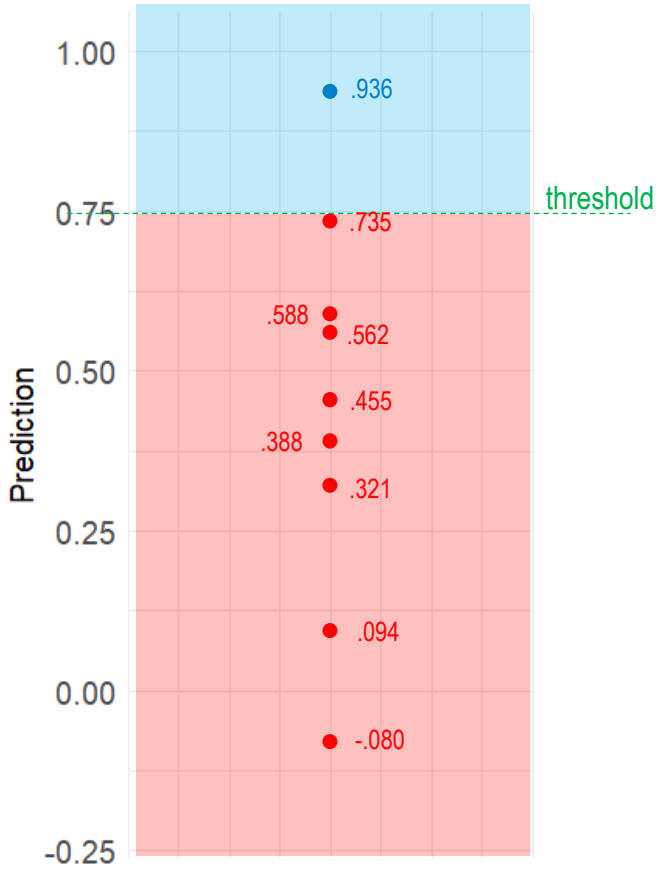
Using Threshold 0.25

		ACTUAL	
		1	0
PREDICTED	1	4	3
	0	0	2

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Using Different Thresholds

X	Y	Pred	Class
-17	0	.094	0
-30	0	-.080	0
0	1	.321	0
18	1	.562	0
31	1	.735	0
46	1	.936	1
10	0	.455	0
5	0	.388	0
20	0	.588	0



Using Threshold 0.75

		ACTUAL	
		1	0
PREDICTED	1	1	0
	0	4	5

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Using Different Thresholds

Using Threshold 0.25

		ACTUAL	
		1	0
PREDICTED	1	4	3
	0	0	2

$$FPR = \frac{FP}{FP + TN} = \frac{3}{3 + 2} = 0.60$$

$$FNR = \frac{FN}{FN + TP} = \frac{0}{0 + 4} = 0.00$$

Using Threshold 0.50

		ACTUAL	
		1	0
PREDICTED	1	3	1
	0	1	4

$$FPR = \frac{FP}{FP + TN} = \frac{1}{1 + 4} = 0.20$$

$$FNR = \frac{FN}{FN + TP} = \frac{1}{1 + 3} = 0.25$$

Using Threshold 0.75

		ACTUAL	
		1	0
PREDICTED	1	1	0
	0	4	5

$$FPR = \frac{FP}{FP + TN} = \frac{0}{0 + 5} = 0.00$$

$$FNR = \frac{FN}{FN + TP} = \frac{4}{4 + 1} = 0.80$$

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